STRUCTURAL HAND CALCULATIONS (AUTOMATED)

Uniform Distributed Load (w) = 15.0 kN/m

Beam Length (L) = 10.0 meters

1. Support Reaction Force (R_A):

R_A = (w * L) / 2 = (15.0 * 10.0) / 2 = 75.0 kN

2. Maximum Bending Moment (M_max):

M_max = (w * L^2) / 8 = (15.0 * 10.0^2) / 8 = 187.5 kNm